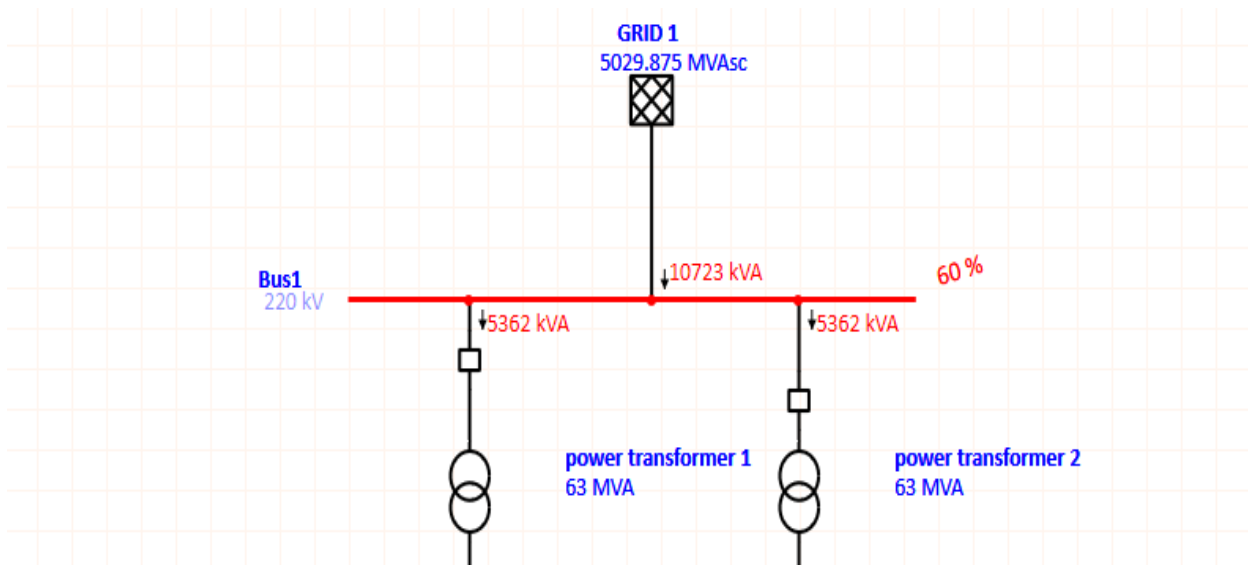
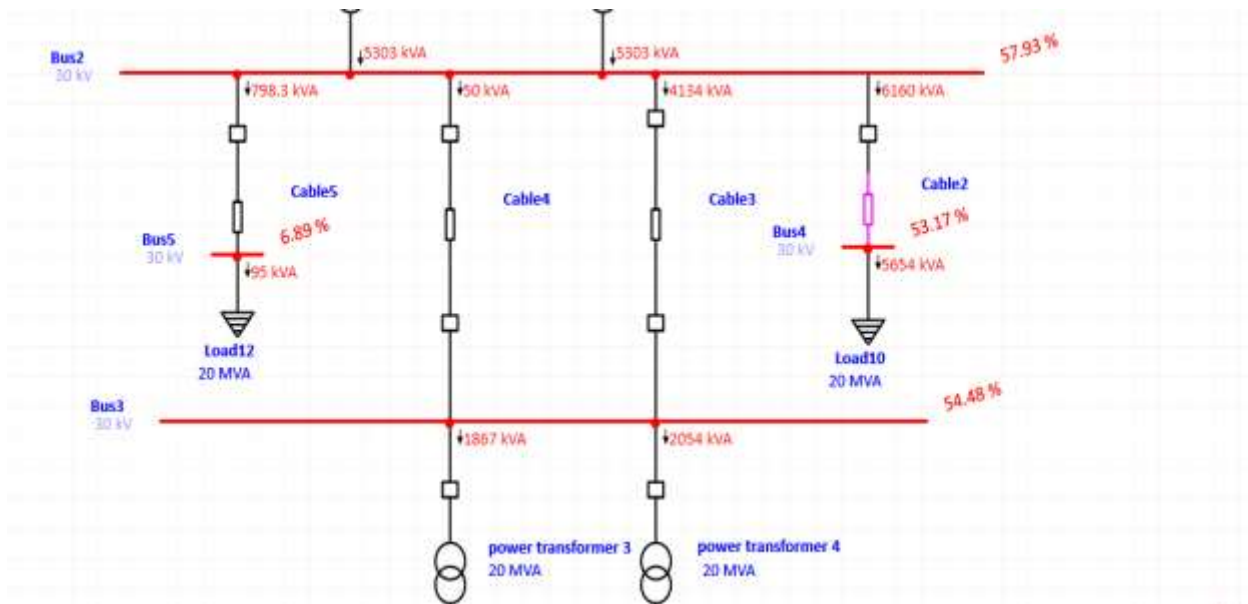


1: Design of schematics

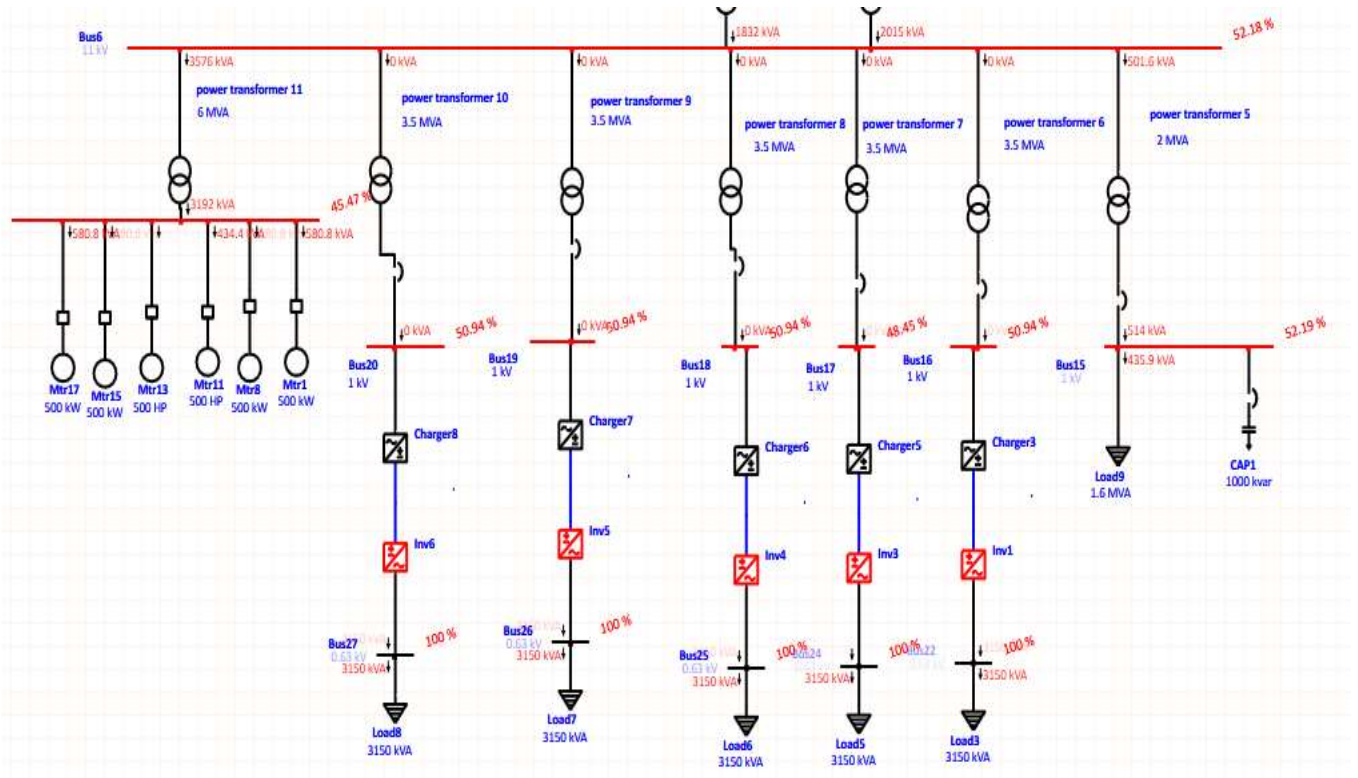
Part(a)



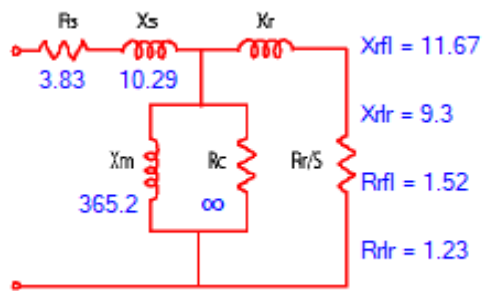
Part(b)



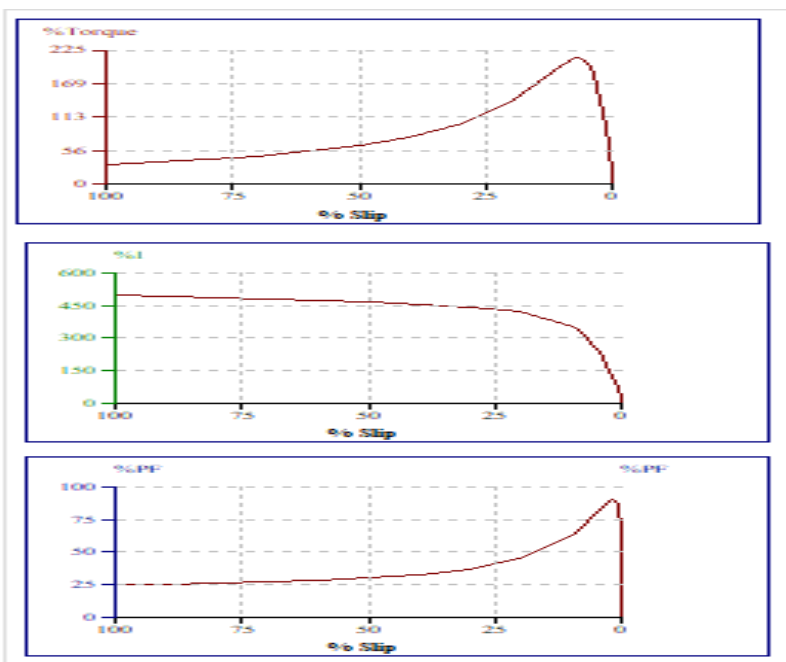
Part(c)



2: Use of Induction motors



Harmonics



* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

Indicates a bus with a load mismatch of more than 0.1 MVA

4: 2-Winding transformer input Data

2-Winding Transformer Input Data														
Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
power transformer 1	3-Phase	63.000	220.000	30.000	12.50	45.00	0	0	10.0	5.000	2.500	13.7500	Dyn	0.000
power transformer 2	3-Phase	63.000	220.000	30.000	12.50	45.00	0	0	10.0	5.000	2.500	13.7500	Dyn	0.000
power transformer 3	3-Phase	20.000	30.000	11.000	10.00	20.00	0	0	10.0	5.000	2.500	11.0000	Dyn	0.000
power transformer 4	3-Phase	20.000	30.000	11.000	10.00	20.00	0	0	0	5.000	2.500	10.0000	Dyn	0.000
power transformer 5	3-Phase	2.000	11.000	1.000	6.25	6.00	0	0	10.0	5.000	2.500	6.8750	Dyn	0.000
power transformer 6	3-Phase	3.500	11.000	1.000	7.15	8.50	0	0	10.0	5.000	2.500	7.8650	Dyn	0.000
power transformer 7	3-Phase	3.500	11.000	1.000	7.15	8.50	0	0	10.0	5.000	-2.500	7.8650	Dyn	0.000
power transformer 8	3-Phase	3.500	11.000	1.000	7.15	8.50	0	0	10.0	5.000	2.500	7.8650	Dyn	0.000
power transformer 9	3-Phase	3.500	11.000	1.000	7.15	8.50	0	0	10.0	5.000	2.500	7.8650	Dyn	0.000
power transformer 10	3-Phase	3.500	11.000	1.000	7.15	8.50	0	0	10.0	5.000	2.500	7.8650	Dyn	0.000
power transformer 11	3-Phase	6.000	11.000	6.000	7.15	8.50	0	0	10.0	5.000	2.500	7.8650	Dyn	0.000

5: Branch losses summary report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Cable2	6.156	0.218	-5.654	0.000	502.1	218.5	57.9	53.2	4.76
Cable3	3.683	1.877	-3.457	-1.779	226.1	98.4	57.9	54.5	3.45
Cable4	0.015	0.048	-0.014	-0.045	1.0	2.8	57.9	54.5	3.45
Cable5	0.360	0.712	-0.095	0.000	265.2	712.5	57.9	6.9	51.04
power transformer 1	5.111	1.620	-5.107	-1.428	4.3	192.1	60.0	57.9	2.07
power transformer 10	0.000	0.000	0.000	0.000	0.0	0.0	52.2	50.9	1.24
power transformer 11	3.021	1.912	-2.942	-1.238	79.3	674.0	52.2	45.5	6.71
power transformer 2	5.111	1.620	-5.107	-1.428	4.3	192.1	60.0	57.9	2.07
power transformer 3	1.653	0.869	-1.649	-0.797	3.6	71.1	54.5	52.2	2.30
power transformer 4	1.818	0.955	-1.814	-0.877	3.9	78.2	54.5	52.2	2.30
power transformer 5	0.442	-0.238	-0.436	0.272	5.8	34.5	52.2	52.2	0.02
power transformer 6	0.000	0.000	0.000	0.000	0.0	0.0	52.2	50.9	1.24
power transformer 7	0.000	0.000	0.000	0.000	0.0	0.0	52.2	48.5	3.73
power transformer 8	0.000	0.000	0.000	0.000	0.0	0.0	52.2	50.9	1.24
power transformer 9	0.000	0.000	0.000	0.000	0.0	0.0	52.2	50.9	1.24
					1095.5	2274.3			

* This Transmission Line includes Series Capacitor.

6:Bus loading summary report

Bus Loading Summary Report													
Bus			Directly Connected Load								Total Bus Load		
			Constant kVA		Constant Z		Constant I		Generic		MVA	% PF	Amp
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar			
Bus22	0.630										3.150	100.0	2886.8
Bus24	0.630										3.150	100.0	2886.8
Bus25	0.630										3.150	100.0	2886.8
Bus26	0.630										3.150	100.0	2886.8
Bus27	0.630										3.150	100.0	2886.8
Bus1	220.000										10.723	95.3	46.9
Bus2	30.000		0.000	0.000							10.605	96.3	352.3
Bus3	30.000										3.920	88.5	138.5
Bus4	30.000			0.000	5.654						5.654	100.0	204.6
Bus5	30.000				0.005						0.005	100.0	26.5
Bus6	11.000			0.000							3.956	87.5	397.9
Bus15	1.000				0.436	-0.272					0.514	84.8	568.6
Bus16	1.000												-
Bus17	1.000												-
Bus18	1.000												-
Bus19	1.000												-
Bus20	1.000												-
Bus21	6.000		2.942	1.238							3.182	92.2	675.5

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).

7: Branch connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
power transformer 1	2W XFMR	Bus1	Bus2	1.45	65.23	65.25	
power transformer 2	2W XFMR	Bus1	Bus2	1.45	65.23	65.25	
power transformer 3	2W XFMR	Bus3	Bus6	8.21	164.22	164.43	
power transformer 4	2W XFMR	Bus3	Bus6	7.46	149.29	149.48	
power transformer 5	2W XFMR	Bus6	Bus15	168.95	1013.69	1027.67	
power transformer 6	2W XFMR	Bus6	Bus16	78.49	667.20	671.80	
power transformer 7	2W XFMR	Bus6	Bus17	74.67	634.65	639.03	
power transformer 8	2W XFMR	Bus6	Bus18	78.49	667.20	671.80	
power transformer 9	2W XFMR	Bus6	Bus19	78.49	667.20	671.80	
power transformer 10	2W XFMR	Bus6	Bus20	78.49	667.20	671.80	
power transformer 11	2W XFMR	Bus6	Bus21	45.79	389.20	391.88	
Cable2	Cable	Bus2	Bus4	123.33	53.67	134.50	
Cable3	Cable	Bus2	Bus3	123.33	53.67	134.50	
Cable4	Cable	Bus2	Bus3	3878.28	10419.71	11118.06	
Cable5	Cable	Bus2	Bus5	3878.28	10419.71	11118.06	

8:Adjustment

Adjustments			
Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		
Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

9: Bus input data

Bus Input Data												
Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	220.000	1	60.0	0.0								
Bus2	30.000	1	100.0	0.0								
Bus3	30.000	1	100.0	0.0								
Bus4	30.000	1	100.0	0.0			20.000	0.000				
Bus5	30.000	1	100.0	0.0			20.000	0.000				
Bus6	11.000	1	100.0	0.0								
Bus15	1.000	1	100.0	0.0			1.600	-1.000				
Bus16	1.000	1	100.0	0.0								
Bus17	1.000	1	100.0	0.0								
Bus18	1.000	1	100.0	0.0								
Bus19	1.000	1	100.0	0.0								
Bus20	1.000	1	100.0	0.0								
Bus21	6.000	1	100.0	0.0	2.942	1.238						
Bus22	0.630	2	100.0	0.0			3.150	0.000				
Bus24	0.630	3	100.0	0.0			3.150	0.000				
Bus25	0.630	4	100.0	0.0			3.150	0.000				
Bus26	0.630	5	100.0	0.0			3.150	0.000				
Bus27	0.630	6	100.0	0.0			3.150	0.000				
Total Number of Buses: 18					2.942	1.238	57.350	-1.000	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus1	220.000	Swing	1	60.0	0.0					
Bus22	0.630	Swing	2	100.0	0.0					
Bus24	0.630	Swing	3	100.0	0.0					
Bus25	0.630	Swing	4	100.0	0.0					
Bus26	0.630	Swing	5	100.0	0.0					
Bus27	0.630	Swing	6	100.0	0.0					
						0.000	0.000			

10: Line/Cable/Busway Input Data

ohms or siemens/1000 ft per Conductor (Cable) or per Phase (Line/Busway)									
Line/Cable/Busway									
ID	Library	Size	Length		#/Phase	T (°C)	R	X	Y
			Adj. (ft)	% Tol.					
Cable2	35NCUN1	1/0	32808.4	0.0	1	75	0.121797	0.053000	
Cable3	35NCUN1	1/0	32808.4	0.0	1	75	0.121797	0.053000	
Cable4			32808.4	0.0	1	75	3.830000	10.290000	
Cable5			32808.4	0.0	1	75	3.830000	10.290000	

Line / Cable / Busway resistances are listed at the specified temperatures.

11: Alert summary report

Alert Summary Report

	<u>% Alert Settings</u>	
	<u>Critical</u>	<u>Marginal</u>
<u>Loading</u>		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
<u>Bus Voltage</u>		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
<u>Generator Excitation</u>		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

12: Critical Report

Critical Report							
Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus1	Bus	Under Voltage	220.000	kV	132.000	60.0	3-Phase
Bus15	Bus	Under Voltage	1.000	kV	0.522	52.2	3-Phase
Bus16	Bus	Under Voltage	1.000	kV	0.509	50.9	3-Phase
Bus17	Bus	Under Voltage	1.000	kV	0.485	48.5	3-Phase
Bus18	Bus	Under Voltage	1.000	kV	0.509	50.9	3-Phase
Bus19	Bus	Under Voltage	1.000	kV	0.509	50.9	3-Phase
Bus2	Bus	Under Voltage	30.000	kV	17.378	57.9	3-Phase
Bus20	Bus	Under Voltage	1.000	kV	0.509	50.9	3-Phase
Bus21	Bus	Under Voltage	6.000	kV	2.728	45.5	3-Phase
Bus3	Bus	Under Voltage	30.000	kV	16.344	54.5	3-Phase
Bus4	Bus	Under Voltage	30.000	kV	15.951	53.2	3-Phase
Bus5	Bus	Under Voltage	30.000	kV	2.067	6.9	3-Phase
Bus6	Bus	Under Voltage	11.000	kV	5.740	52.2	3-Phase
Inv1	Inverter	Overload	8.248	Amp	2886.752	35000.0	3-Phase
Inv3	Inverter	Overload	8.248	Amp	2886.752	35000.0	3-Phase

13: Branch loading summary report

CKT / Branch		Busway / Cable & Reactor			Transformer				
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	Loading (input)		Loading (output)	
						MVA	%	MVA	%
Cable2	Cable	211.66	204.65	96.69					
Cable3	Cable	211.66	137.33	64.88					
power transformer 1	Transformer				63.000	5.362	8.5	5.303	8.4
power transformer 2	Transformer				63.000	5.362	8.5	5.303	8.4
power transformer 3	Transformer				20.000	1.867	9.3	1.832	9.2
power transformer 4	Transformer				20.000	2.054	10.3	2.015	10.1
power transformer 5	Transformer				2.000	0.514	25.7	0.502	25.1
power transformer 6	Transformer				3.500	0.000	0.0	0.000	0.0
power transformer 7	Transformer				3.500	0.000	0.0	0.000	0.0
power transformer 8	Transformer				3.500	0.000	0.0	0.000	0.0
power transformer 9	Transformer				3.500	0.000	0.0	0.000	0.0
power transformer 10	Transformer				3.500	0.000	0.0	0.000	0.0
power transformer 11	Transformer				6.000	3.576	59.6	3.192	53.2

* Indicates a branch with operating load exceeding the branch capability.